Stability and finite starting cases for symmetric string C-groups

Human Compression project

Abstract

We recover the principal results of Cameron, Fernandes and Leemans, *The number of string C-groups of high rank*, in its arXiv version of 11 January 2023. The rank and degree correspondence is proved using pure perfect cuts, actual subgroup intersections and an explicit contraction. A distinguished pair of pure cuts reduces the six finite starting counts to rooted markings on at most twelve points. The resulting independent enumeration gives the original constants $1, 1, 7, 9, 35, 48$. The complete proofs of the two acquired components and their supporting theory accompany this assembly; numerical constants are not borrowed from the original table.

A string C-marking is an ordered list of nonidentity involutions $\rho_0, \dots, \rho_{r-1}$ satisfying distant commutation and the intersection property

$$\langle \rho_i : i \in I \rangle \cap \langle \rho_i : i \in J \rangle = \langle \rho_i : i \in I \cap J \rangle$$

for all label sets I, J. Marked-group isomorphisms preserve the ordered marks. Reversal of their order is duality. Let $\Sigma^\kappa(n)$ be the markings of $\mathrm{Sym}(n)$ of rank $n - \kappa$, modulo marked-group isomorphism and duality. This is the convention of the source's introduction and Section 2.

Theorem 1 (The principal stability statement)**.** *For every $\kappa \geq 1$ and $n \geq 2\kappa + 3$, there is a bijection*

$$\Sigma^\kappa(n) \longrightarrow \Sigma^\kappa(n+1).$$

It is obtained by subdividing an internal pure perfect cut. Its inverse contracts the first internal pure pair. Both maps are well defined on the stated isomorphism and duality classes. Thus every class at a later degree is obtained from a class at the boundary degree $2\kappa + 3$.

Proof. Apply the complete acquired component *Pure cuts and the stable counting correspondence*, Theorem 7. We record its exact interfaces to make the application explicit. A pure perfect cut has mark $(a\ b)$, and its preceding and following intervals are supported and transitive on the two omission orbits containing a, b. For a fractured symmetric marking with $2r > n+2$, the acquired sufficient-split induction and the full leaf-kernel criterion give an internal pure perfect cut. Fracture is automatic in the required range for $n \geq 7$. Degrees five and six use the ordinary simplex representative, with the exceptional outer action of S_6 identified only at the level of abstract marked groups.

The acquired proof shows that all pure perfect cuts form a single consecutive path, with every internal path vertex fixed by the other marks. Subdividing a pure edge inserts one point and one mark. Actual one-bridge subgroup intersections prove that contracting an adjacent pure pair preserves IP and generates the full symmetric group on one fewer point. No quotient-IP inference is used. Exact omission orbit transport shows that the two operations are inverse at the least internal pure cut. The density inequality makes that cut begin an internal pure pair in the target. Subdivision at any internal position of the pure path gives an ordered-conjugate marking; this also pays compatibility with reversal, including self-dual classes.

For the present parameters, $r = n - \kappa$ and $2r \geq n + 3$, so all domain hypotheses hold. In the target, $N = n + 1$, $R = r + 1$, and $2R \geq N + 4$, precisely the range used for contraction. The maps therefore have the displayed domains and codomains. Iteration gives the stated constructive reduction. □

Theorem 2 (The six finite starting counts)**.** *Writing* $c_\kappa = |\Sigma^\kappa(2\kappa+3)|$*, one has*

$$(c_1, \ldots, c_6) = (1, 1, 7, 9, 35, 48).$$

These are the values of $|\Sigma^\kappa(n)|$ *for every* $n \geq 2\kappa + 3$ *and* $1 \leq \kappa \leq 6$*.*

Proof. The complete acquired component *The finite starting cases from balanced pure cuts*, Theorem 1, gives a unique consecutive pure pair whose prefix surpluses are zero and one at the boundary $n = 2r - 3$. Removing its middle vertex leaves two rooted transitive fractured C-markings of degrees $2p, 2q$ and ranks p, q, with $p + q = r - 2$. Each admits a pure leaf at its root. Conversely the exact leaf criteria, the actual bridge intersection identity and pure subdivision reconstruct the boundary marking. Ordered point conjugacy gives precisely rooted point conjugacy on the two wings, and reversal exchanges the wings.

Theorem 3 of that component computes the rooted wing numbers by a complete literal involution search, giving $(w_1, \ldots, w_6) = (1, 1, 6, 3, 11, 19)$. All first involutions are considered; the search is over actual permutation tuples modulo the root stabilizer, including degree-six actions without identifying them by outer group automorphisms. The completeness proof, all representatives, terminal receipts, independent labelled controls and actual IP checks are retained. The unordered-pair formula consequently gives

$$c_\kappa = \sum_{\substack{p<q \\ p+q=\kappa+1}} w_p w_q + \begin{cases} \binom{w_h+1}{2}, & \kappa + 1 = 2h, \\ 0, & \text{otherwise.} \end{cases}$$

This yields respectively $1, 1, 6 + 1, 3 + 6, 11 + 3 + 21, 19 + 11 + 18$. Theorem 1 transfers these boundary values to all degrees in the stated ranges. □

Source contract and attribution. Theorem 1 recovers source Theorems 1.1 and 9.10 and the constructive classification reduction stated after Theorem 1.1. Theorem 2 recovers the six constants in the abstract, introduction and the relevant boundary entries of Table 1. These were already results of the original authors; no new resolution of an open conjecture is claimed. Other entries of the historical table are not silently added to this principal contract.

The original rank and degree operation, the rank classifications and the established C-group criteria retain their original attribution. The new proof uses backward propagation of full leaf kernels, a single interval of pure cuts, explicit actual contraction and a balanced pair for the finite count. The supporting recovered components distinguish these deductions from reused proofs and computational inputs. The prior accepted corrections to the unrestricted converse in source Proposition 2.4, the indicated case of Proposition 2.5, and the unrestricted extension claim in Theorem 6.4 remain in force. The proof above uses only their exact repaired interfaces; it does not restore the false unqualified statements. Those auxiliary counterexamples do not refute the principal stability theorem.

The source edition is arXiv:2212.12723v2, 11 January 2023. A later edition and the OEIS entry contain an additional numerical term; it is outside this fixed edition's six-term contract and is not inferred here. The separately registered classification paper L042 keeps its own principal contract and original count.